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SUBIR CHOWDHURY SCHOOL OF QUALITY & RELIABILITY



RELIABILITY DEMONSTRATION LAB REPORT
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RELIABILITY DEMONSTRATION OF PUSH BUTTON SWITCHES

Submitted By

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INDEX

Ser	Description	Page No
I	List of Figures	4
ii	List of Tables	5
1	Objective	6
2	Introduction	6
3	Push Button Switch Theory & Operation	6
4	Test Operating Conditions	7
5	Test Setup	8
6	Test Procedure	9
7	Test Observations	13
8	Inference	14

LIST OF FIGURES

Fig No	Title	Page No
1	Push Button Switch	7
2	Circuit Diagram of the Test Set up	8
3	Actual Test Set up	9
4	Sequential Testing Plot	12

LIST OF TABLES

Table No	Title	Page No
1	Switch Specifications	7
2	Switch Operating Conditions	7
3	Acceptance & Rejections Boundary Values for Sequential Test	12
4	Sequential Test Parameters and Minimum Sample Size Calculation	13
5	Test Results	13-14

Objective

1. To demonstrate the minimum acceptable reliability of push button switches for a specified operating time with given Producer's and Consumer's Risk.

Introduction

2. Reliability demonstration is a statistical testing method used to verify whether a product meets a specified reliability requirement. In our experiment, we used a binomial sequential test plan to demonstrate that the push button switch Reliability is at least 0.95 for 2683 cycles under given operating conditions for a given Producer's Risk = 0.2 and Consumer's Risk = 0.2. The main factors to be considered when planning and implementing Reliability Demonstration are as listed below:-

- 2.1. **Define.** Define the R_U (Desired or claimed reliability level), R_L (Minimum acceptable reliability level)
- 2.2. **Risk.** Set acceptable statistical error margins (Producer's risk, Consumer's risk).
- 2.3. **Scale.** Determine the required sample size and total test time.
- 2.4. **Environment.** Choose between normal operating conditions or accelerated testing.
- 2.5. **Method.** Pick the test design (Sequential).

Push Button Switch Theory & Operation

3. A push-button switch is a simple switch mechanism for controlling some aspect of a machine or a process. Buttons are typically made from hard material, usually plastic or metal. The surface is usually flat or shaped to accommodate the human finger or hand, to be easily pressed or pushed. Buttons are most often biased switches, though even many unbiased buttons (due to their physical nature) require a spring to return to their unpushed state. The push-button switch used in the experiment is a momentary or non-latching switch which causes a temporary change in the state of an electrical circuit only while the switch is physically actuated.

4. **Construction and Operation.** The push button switch (**Fig 1**) used in the experiment is normally-open Single Pole Single Throw (SPST) momentary switch, wherein an automatic mechanism (i.e. a spring) returns the switch to its default position immediately after releasing the button, therefore restoring it to its initial circuit condition. There are two types of push button switches:-

- 4.1. **Normally Open.** Pushing the switch allows electricity to flow between its two contacts when held in. When the button is released, the circuit is broken.

4.2. **Normally Closed.** Pushing the switch does the opposite, i.e. when the button is not pressed, electricity can flow, but when it is pressed the circuit is broken.

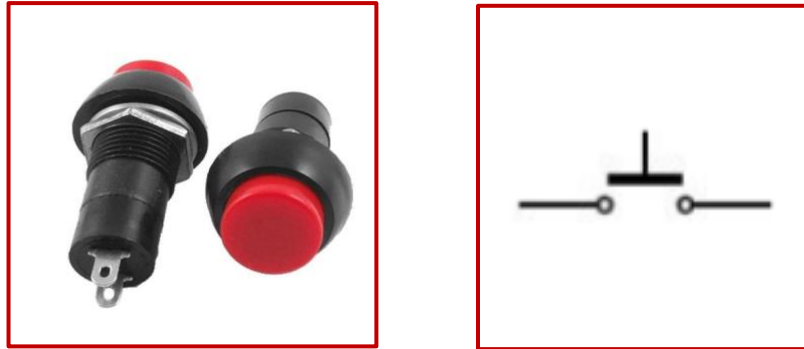


Fig 1 - Push Button Switch

5. **Specifications.** Push Button Switch (SE952 Push To On Medium) has the following technical specifications (**Table 1**):-

Switch Type	Push Button Momentary, Round head
Terminal Style	Screw type
Operating Temp.	10°C to 80°C
Mounting	Panel, screw type
Current & Voltage Rating	5A, 220 V AC

Table 1 - Switch Specifications

Test Operating Conditions

6. The operating conditions of this experiment are as follows (**Table 2**):-

Electrical Supply	220 V AC supply
Temperature	25 °C
Cycles per minute	30
Relative Humidity	70 – 80 % (approx.)

Table 2 - Switch Operating Conditions

7. Reliability Testing Using Inverse Power Law Model. The course had performed Reliability Testing of push button switches using accelerated life testing at stress levels of **0.8 A, 1 A, and 1.5 A** with 10 samples at each level in Autumn term 2025-26. The failure data was analyzed using the **Weibull distribution**, and the **Inverse Power Law (IPL)** model was applied to relate stress and life. From the analysis the following final equation was obtained for the reliability of the push button switch:-

$$R(t) = e^{-\left(\frac{t}{\left(\frac{A}{S^B}\right)}\right)^\beta}$$

7.1. Weibull shape parameter $\beta=3.03677$, indicating **wear-out failure behavior**.

7.2. Inverse Power model constants were determined as:-

7.2.1. $A=10153.77$

7.2.2. $B=1.7414$

7.3. Using this model, the life at **nominal stress (0.2 A)** was estimated as $\eta=167422$ cycles.

7.4. The reliability at the mission time of **30,000 cycles** was calculated as **$R_u=0.99$** .

7.5. The minimum acceptable reliability level was assumed as **$R_L=0.95$** .

Test Setup

8. In the experiment, an open type of push-button switch was used. The circuit was connected as shown in the circuit diagram below (**Fig 2**). The circuit is designed in such a way that the failure of the switches can be accelerated by either increasing the rate of switching speed (on - off cycle) in rated load conditions or by increasing the load. Image of actual test setup is presented as **Fig 3**.

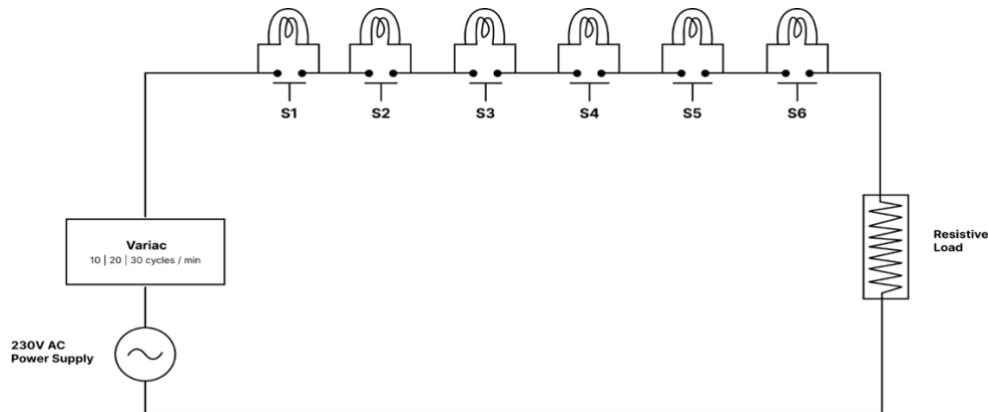


Fig 2 - Circuit Diagram of the Test Setup



Fig 3 - Actual Test Setup

9. **Failure Criteria.** Failure is defined as the point at which the system or component can no longer perform its intended function. The following are the 3 major failure modes for the push buttons:-

- 9.1. **Open Circuit.** No current flow when the button is pressed.
- 9.2. **Short Circuit.** Continuous current flow when the button is released.
- 9.3. **Mechanical Failure.** Spring dislodgement / mechanical wear.

Test Procedure

10. A Binomial Sequential Test Plan is selected for this analysis due to its efficiency in evaluating the system's reliability by minimizing the average sample size while maintaining statistical rigor.

11. **Sequential Testing.** The term **sequential** indicates that this type of test is accomplished by conducting a succession, or a **sequence** of runs on one or more devices. Test runs are continued on each device in the test sample until failure occurs, except when an **accept** decision is reached, till limit of the operating life specified for the device. The specific practice that will be followed in the event of a failure; that is, repair or replacement of the failed test device, is decided by the engineer establishing the requirements for the test and does not affect the statistical validity of the test results.

12. One of the following three decisions is made after each test run:-

- 12.1. **Accept.** The device has demonstrated the reliability requirement.
- 12.2. **Continue Testing.** The status of the device's reliability is undetermined.

12.3. **Reject.** The device has failed to demonstrate the reliability requirement.

12.4. Testing is terminated if either Decision 12.1 or 12.3 is reached.

13. The following procedure was adopted for the experiment:-

13.1. At a time, six push buttons were connected in series along with an incandescent bulb parallel to each switch for indication of the serviceability of the switch.

13.2. The switches were subjected to continuous ON/OFF cycles using a motor driven cam and pinion mechanism. The speed was set at 30 cycles per minute for all batches. Load current of 0.8A was chosen. The number of cycles to failure was continuously monitored using a counter.

13.3. Since the switches used were normally open, pressing them would close the circuit, causing less current to pass through the parallel-connected lamp, which would remain off. When the spring retracted, more current would flow through the lamp, causing it to glow.

13.4. If a switch failed in short-circuit mode, the lamp would stop glowing due to the continuous flow of current through the shorted switch.

13.5. If a switch failed in open-circuit mode, the corresponding lamp would remain on, as the current would always flow through the lamp.

13.6. Upon any failure of the switches (either short-circuit or open-circuit), the switch was considered failed and was not replaced with a new one. The test would be then restarted. Failures due to short-circuit or open-circuit conditions were the accepted mode of failure for the test.

13.7. The test was conducted one batch at a time, consisting of six push-button switches. A total of 6 batches were tested.

14. **Values Considered (Assumptions).**

14.1. R_D = Design objective for the reliability level = 0.99

14.2. R_L = Minimum acceptable reliability level to the customer = 0.95

14.3. α = Producer's risk = Rejecting item with true reliability = 0.2

14.4. β = Consumer's risk = Accepting item with true reliability = 0.2

14.5. $Q_U = 1 - R_L = 0.05$

14.6. $Q_L = 1 - R_U = 0.01$

14.7. The consumer's risk, β , is associated with the minimum acceptable reliability, R_L .

14.8. The producer's risk α , is associated with the design objective reliability, R_U .

15. **Construction of Binomial Sequential Test Plan.** The following step-by-step method may be used to construct a **binomial sequential testing plan**, where it is necessary to demonstrate that a device meets or fails to meet a specified, acceptable minimum reliability, R_L :-

15.1. Determine the value of the accept intercept
$$h_0 = \frac{\ln\left(\frac{\beta}{1-\alpha}\right)}{\ln\left(\frac{Q_U}{Q_L}\right) - \ln\left(\frac{R_L}{R_U}\right)}$$

15.2. Determine the value of the reject intercept
$$h_1 = \frac{\ln\left(\frac{1-\beta}{\alpha}\right)}{\ln\left(\frac{Q_U}{Q_L}\right) - \ln\left(\frac{R_L}{R_U}\right)}$$

15.3. Determine the value of the slope of the test plan lines.
$$S = \frac{\ln\left(\frac{R_U}{R_L}\right)}{\ln\left(\frac{Q_U}{Q_L}\right) - \ln\left(\frac{R_L}{R_U}\right)}$$

15.4. Determine the equations for the accept line, $AL_f = h_0 + SN$

15.5. Determine the equations for the reject line, $RL_f = h_1 + SN$

15.6. **Maximum achievable Reliability (R_U):** $R_U = 99\%$ at 30,000 cycles life under normal operating conditions (30 cycle per minute).

15.7. **Minimum acceptable reliability (R_L):** $R_L = 95\%$ at 30,000 cycles life under normal operating conditions (30 cycle per minute).

15.8. **Calculated Parameters.** The parameters required for construction of Sequential Test plan is as below. Plot of the plan indicating the no. of failures vs trials is depicted in **Fig 4** below. The blue line in Fig 4 is indicating **Reject Line** and orange line is indicating **Accept Line**.

Accept line intercept: -0.839832

Reject line intercept: 0.839832

Slope: 0.024985

Minimum No of Trials: 34 samples

$$AL_f = -0.839832 + 0.024985N$$

$$RL_f = 0.839832 + 0.024985N$$

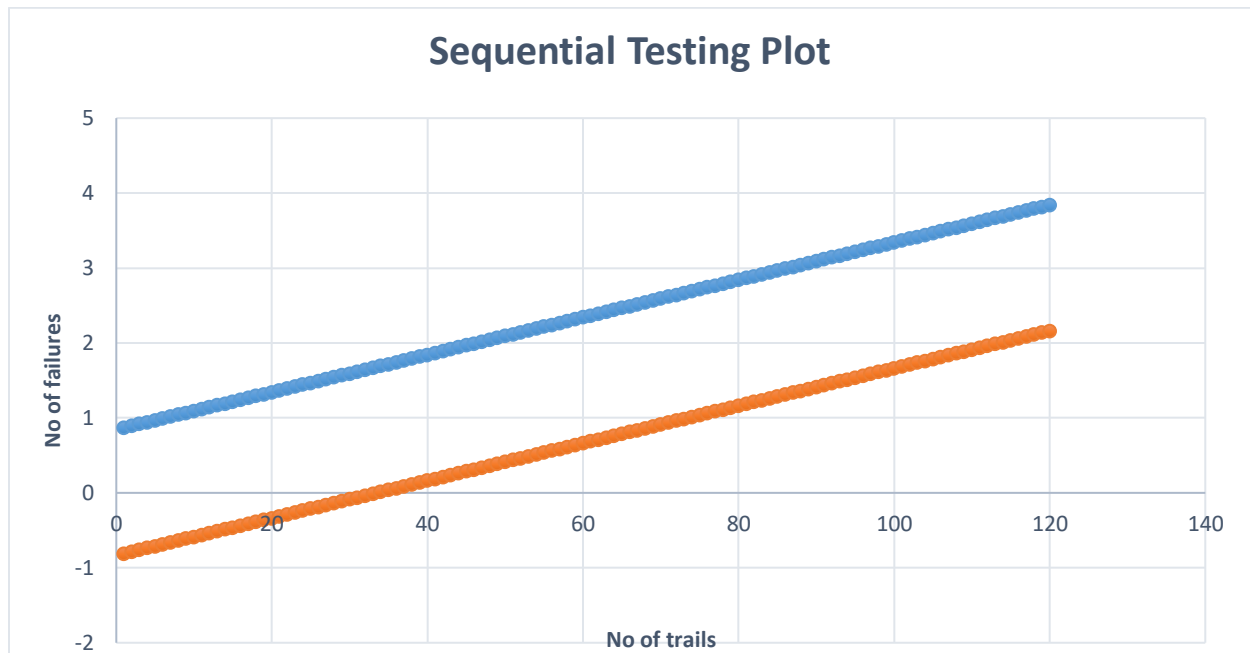


Fig 4 - Sequential Testing Plot (Acceptance & Rejection Lines)

15.9. **Table 3** below indicates Accept and Reject boundaries for given sample sizes.

Number of Trials	Reject Line	Accept Line
1	0.864817	-0.814846
2	0.889803	-0.789861
3	0.914788	-0.764876
4	0.939774	-0.73989
5	0.964759	-0.714905
...	...	...
34	1.689336	0.009673
...	...	...
114	3.68817	2.008506
115	3.713155	2.033492
116	3.738141	2.058477
117	3.763126	2.083463
118	3.788112	2.108448
119	3.813097	2.133433
120	3.838083	2.158419

Table 3 – Acceptance & Rejection Boundary Values for Sequential Test

16. **Test Duration.** A total of 34 push button samples were divided into six groups (6, 6, 6, 6, 6, and 4). Each sample was tested for more than 2683 cycles under specified conditions and observed for any failure occurring within 2683 cycles. The test duration of 2683 cycles were determined using the electrical life of 30,000 cycles and the Acceleration Factor (AF = 11.18) at a stress level of 0.8 A,

$$\text{Test cycles} = \text{Electrical life} / \text{AF} = 30,000 / 11.18 \approx 2683 \text{ cycles}$$

17. **Table 4** brings out the minimum number of samples (assuming 0 failures) required to be tested for a given Producer's Risk and Consumer's Risk:-

beta	alpha	h0	h1	S	MIN SAMPLE
0.1	0.1	-1.3311	1.331102	0.024985	53
0.15	0.15	-1.05084	1.05084	0.024985	42
0.2	0.2	-0.83983	0.839832	0.024985	34

Table 4 - Sequential Test Parameters and Minimum Sample Size Calculation

Test Observations

18. A total of 34 push button samples were tested at 0.8 A for 2683 cycles. It was observed that no failures occurred during the test. All samples successfully functioned during the required test cycles and satisfied the sequential test acceptance criteria. Hence, the push buttons are considered to meet the specified reliability requirements.

19. **Test Results.** Test results of 34 push buttons are brought out in **Table 5** below.

Push Button Ser	Status	Batch
1	All push buttons operated satisfactorily upto 2683 cycles.	1
2		
3		
4		
5		
6		
7		2
8		
9		
10		
11		
12		
13		3
14		
15		
16		

Push Button Ser	Status	Batch
17	All push buttons operated satisfactorily upto 2683 cycles.	
18		
19		4
20		
21		
22		
23		
24		
25		5
26		
27		
28		
29		
30		
31		6
32		
33		
34		

Table 5 – Test Results**Inference**

20. From the sequential testing plot, it is observed that the acceptance line intersects the trials axis at 34, indicating that acceptance is achieved with 0 failures up to 34 samples.

21. The absence of failures during the test indicates that the push buttons exhibit high reliability under the given operating conditions.

22. The product satisfies the required reliability limits ($R_L = 0.95$) with the specified producer's and consumer's risk levels.

23. The sequential testing method efficiently validates reliability using a minimum number of samples, thereby reducing testing time.

24. Hence, the push buttons are considered reliable and suitable for practical applications within the specified limits.

END OF REPORT